Shaishav Shah

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Experience

Palantir Technologies **New York, US**
Software Engineer *Aug 2024 – Present*

- Diagnosed duplicate execution and database contention; independently built **distributed concurrency libraries** with service-scoped work ownership, optimizing deletion by **10×+**.
- Designed **heartbeat-based leases, TTL expiration, and automated work reclamation** for node-failure recovery; owned architecture and rollout across shared libraries and services without downstream integration changes.
- Designed and built a **Java security-policy service** leveraging just-in-time **prefix tries**, reducing evaluation time for comparable requests from **∼3 minutes to ∼200 ms** (∼900× faster).
- Developed an **agentic knowledge marketplace** spanning incident ingestion, skill synthesis, updates, conflict detection, and pruning, enabling a **continuously maintained contextual-knowledge base** used across agentic workflows and standalone agents, reducing token-spend by 15%.
- Partnered with security stakeholders to deliver **self-service passkey recovery** with identity verification and auditable event logs, reducing recovery turnaround from **weeks to minutes**.
- **Mentored two engineers** through structured onboarding, technical guidance, and design reviews, enabling both to **join the team's support rotation within 1.5 months**.

Software Engineer Intern *May 2023 – Aug 2023*

- Designed and executed a **zero-downtime migration** from monolithic user permissions to **fine-grained access controls**, enabling scalable role-based authorization.
- Optimized **Java** services through caching, proxy objects, and multithreading, reducing orchestration endpoint **p99 latency by over 41%**.
- Engineered performant, lightweight, and **modular pages** using **TypeScript** and **React**, architecting **dynamic imports** to reduce initial load times by **14%**.

Niricson Software Inc. **Victoria, Canada**
Software Engineer Intern *May 2022 – Aug 2022*

- Automated **cloud GPU provisioning and teardown** for machine learning workflows in **Python**, saving **$17,000 annually**.
- Designed a **geospatial vector data format** for **Amazon DocumentDB** that stored over **50% more data points** and reduced loading time by **80%**.
- Created a custom **GeoJSON polygon simplification algorithm** that reduced memory usage and scaled real-time rendering capacity from **1K to 500K+ data objects**.

Projects

Oasis (Stanford University - TreeHacks 2024 - Winner) **Feb 2024**

- Built a **retrieval-augmented generation (RAG)** system with **LangChain** and vector databases to query and compare large, unstructured ESG documents.
- Fine-tuned **Llama 2** and used **SentenceTransformers** for semantic alignment, extracting and structuring metrics for indexing, retrieval, and cross-company benchmarking.

Technical Skills

Languages: C++, Java, C, Python, TypeScript, JavaScript, SQL, HTML, CSS

Technologies: Spring, NoSQL, Node.js, React, Redux, Pandas, Boost C++, Android

Infrastructure: AWS EC2, SES/SNS, S3, RDS, EFS, Google Cloud, Azure, Firebase

AI Tools & Practices: Codex, Claude Code, Agents API, Decisions API, MCP, Subagents, Knowledge Bases, AI Marketplaces, Operational-Layer AI, Spec-Driven Development

AI Frameworks: Foundry, LangChain, RAG, Llama 2, SentenceTransformers

Education

University of Alberta **Sep. 2020 – May 2024**

Specialization in Computing Science | Business Minor *Edmonton, Alberta*